

A pre-feasibility study for the establishment of a factory for the manufacturing of glucose in Mafraq



2017



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1. Executive Summary

The idea of the project is summarized in manufacturing raw glucose by decomposing the starch by chemical means and sequential operations to achieve the high purity according to the specifications and standards used for medicines in order to be used in the medical and food industries.

The idea was adopted because of the availability of manpower in the province who are able to work in addition to the high profitability of such projects. It is worth mentioning that to ensure the success of the project, it is necessary to know the requirements of domestic and foreign markets.

Table 1 Economic Indicators

Indicator	
Total Investment Cost (Jordan Dinars)	1,443,253
Internal Rate of Return (IRR)	25.34%
Pay-Back Period (Years)	5
Net Present Value (Jordan Dinars)	2,243,038
Job Opportunities provided by the Project	30
Total Revenues (For 10 Years)	12,557,098
Total Operating Expenses (For 10 years)	6,565,065
Total Net Profit (For 10 years)	2,038,619

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

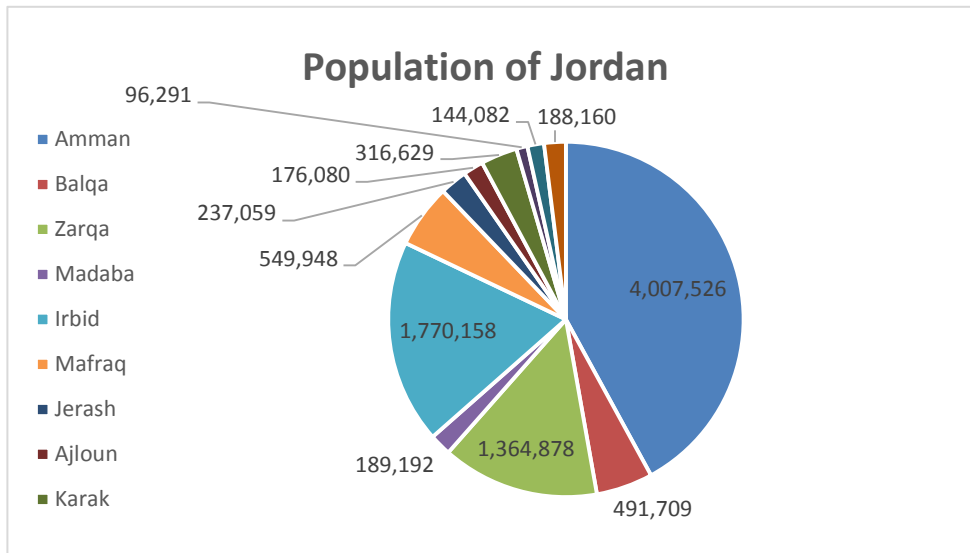
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

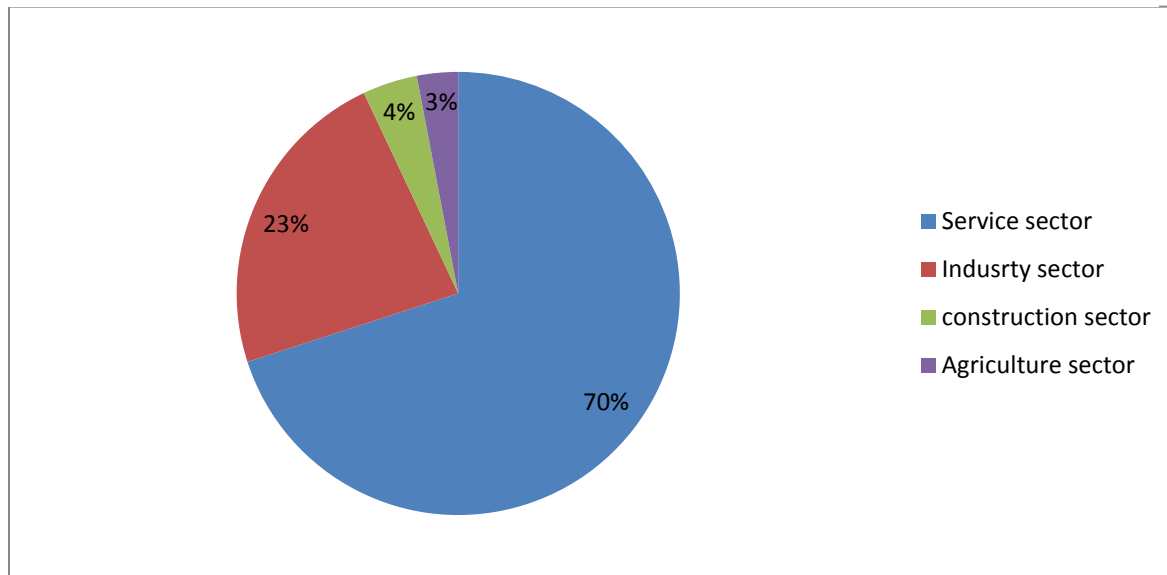
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally. The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Mafrq:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 Payable Income Tax and Tax Exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Mafrq governorate is classified as category C where investors get 60% tax exemption on income tax for 20 years from the date of investment.

2. Market Analysis

Project Description

The idea of the project is summarized in manufacturing raw glucose by decomposing of starch by chemical means and sequential operations to achieve the high purity according to the specifications and standards used for medicines in order to be used it in the medical and food industries.

The idea was adopted because of the availability manpower in the province who are able to work in addition to the high profitability of such projects

Project Objectives

Project objectives are summarized in the following points:

- Achieve profitable returns for the project owners.
- Production of raw glucose (dextrose) in a common form and filling it in appropriate containers.
- Employing local workers and reducing unemployment.
- Provide the targeted markets with their needs of raw glucose.
- manufactured in powder form packed in special cans for its uses as food.

Proposed Product

Provision of raw glucose (dextrose) to be produced and distributed to the targeted segments to achieve the final product of intravenous fluids or food.

Target Segments

- Local factories specialized in producing intravenous solutions.
- Local manufactures specialized in producing food.

Market Size

The project market depends on the category of factories producing medical materials and food-producing factories.

The proposed location for this project is in Al Mafrq, inside the industrial area. It emerged from the preliminary study of the market that intravenous fluids (glucose) are not produced in Jordan. There are five hospitals in AlMafrq, the following table shows the amounts of the annual consumption of glucose solutions and these solutions concentrations are (2.5%, 5%, 10%, 25%) and in different sizes (1 liter, 500 ml). Depending on the information from existing hospitals in various governorates, the amount of consumption of the solutions with concentration of 25%, compared to the others is the lowest. As well as the consumption of glucose solution rate per capita is 2 liters per year. The following table shows the amount of the annual consumption of intravenous fluids (glucose) in hospitals in AlMafrq:

Table 3 annual consumption of glucose solutions in hospitals

Hospital Name	Consumption per year
Mafrq	-
Ruwaished	600 bottles
Obstetrics and Gynecology \ Children	4080 bottles
Badia	4080 bottles
Al Noor Sanatorium	60 bottles

Source: Medical warehouses

Jordan has seven suppliers approved by the Food and Drug Administration for the relevant solutions.

1.1 The Analysis of Competitors

This is the first project of its kind, so there is no direct competitors for the project in the Kingdom but there are suppliers who import glucose solutions.

Production Capacity

The production capacity of the raw glucose project of 517 kg / h was calculated using 300 working days per year and 8 working hours per day

Production capacity = $517 \times 8 \times 300 = 1240.8$ tons \ years.

	Quantity (tons)	%
Dextrose used in food	279.18	20.0%

Dextrose used in intravenous fluids	1116.72	80.0%
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	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%

Price Analysis and Pricing Policy

Table 4 Cost of Raw Materials

Required raw materials	Quantity	Cost (JD)
Dextrin	1 ton	2500
Preservatives	1 ton	1700

According to the prices of improters

Table 5 Products Prices

Product	Quantity	Price (JD)
Dextrose used in food	1 ton	550
Dextrose used in intravenous fluids	1 ton	1600

- The prices of some of the competing companies were approved from outside Jordan

Projected revenues

Table 6 project revenues during the years 2018 to 2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	40%	45%	50%	55%	60%	65%	70%	75%	80%	85%
Revenue of dextrose used for food	67003.2	77640.0	86266.6	94893.3	103519.9	112146.6	120773.3	129399.9	138026.6	146653.3
Revenue of dextrose for intravenous fluids	714700.8	828159.6	920177.3	1012195.0	1104212.7	1196230.5	1288248.2	1380265.9	1472283.6	1564301.4
Total revenue	781704	905800	1006444	1107088	1207733	1308377	1409021	1509666	1610310	1710955

3. Technical Analysis

Required Human Resources

Table 7 Required Labor

Job Title	No.
Indirect staff	
General manager	1
Production Manager	1
Financial Manager	1
Marketing manager	1
Human Resource Manager	1
Secretaries	1
Accountant	2
Salesman	2
Human Resources Officer	1
Guard	1
Cleaner	1
Total Non-Direct staff	13
Technicians	3
Workers	12
Driver	2
Total Direct staff	17
Grand Total	30

Job Description:

General Manager: Proposes the general policy of the plant, prepare special plans and follow up their implementation, supervise the technical and administrative staff, and develop procedures and ensure the requirements of occupational safety and health.

Financial Manager: Implements all the plans, regulations, administrative, accounting and financial procedures, as well as monitoring and controlling the resources and expenditures of the department, and the organization of the budgetary and activities in the department. Also, prepare financial reports.

Production Manager: Responsible for the technical and production affairs, oversees the production process, and ensure the requirements of occupational safety and health

Human Resources: Planning, recruitment of human resources for the company and work to measure their performance and develop their abilities through effective training programs and work to stimulate these cadres on creativity and innovation in the areas of their business and work to regulate the accumulation of experience and knowledge in the areas of work.

Laboratory technician: Responsible for the organization of work within the laboratories and ensure the required quality.

Marketing staff: Study the market needs, and follow-up with advertising, media and assess the effectiveness of the activities, and monitor the level of sales.

Accountant: Responsible for the implementation of financial and accounting procedures. Financial audit in accordance with applicable regulations, in the plant.

Sterilization technician: Responsible for sterilizing the produced container and tools used, where the outputs of the production process to be sterilized in high quality.

Packaging technician: Responsible for packaging and arranging products in sterilized packages and prepare them for selling.

Guard: Maintaining the security of the plant.

Maintenance Technician: Implement actions and activities of the electrical and mechanical maintenance, and follow up the implementation of plans and programs planned in order to ensure the continuity of the production process and quality at full capacity with the lowest possible cost, and make sure of the availability of spare parts, and the preparation of technical reports.

Secretary: Answer phone calls and arranging meetings for managers.

Public Safety staff: Study the potential risks in the factory and work to provide a safe working environment for risk-free.

Development Department staff: Develops efficiency and product quality and reduce costs of production.

Quality department staff: Quality control of the product by reducing defects, workmanship and improve production.

Cleanliness labor: Maintain the cleanliness of the plant and offices.

Driver: Deliver products to customers.

Site Analysis

The factory will be established inside the industrial zone in AlMafrq, and proposed locations for this project, Al Thuria industrial zone, which is characterized by a strategic location in AlMafrq because it lies at the crossroads of international roads linking Jordan with neighboring countries, and possesses a safe investment environment under the laws and investment regulations in Jordan which aims to strengthen the incentives granted to all investors and simplify the creation of project's procedures and obtaining the necessary licenses to operate and benefit from the exemption of customs duties and the equal treatment of the foreign investor in Jordan to the treatment of the Jordanian investor, as well as easy access to the Arab and international markets.

Technical Requirements Analysis

1.1.1 Land and Construction

Since the laws of the industrial zone in Al Mafrq state that the lowest space for the plant must be 5 Donms, including 2 Donms for construction works. The project will be constructed on an area of 5000 sq.m. The following table shows the needed land:

Table8 Required Land for The Project

Land	Cost (JD/sq.m)	Area sq.m
The required land	25	5,000

Source: Land registration – Al Mafrq

The following table shows the required construction work for the project:

Table 9 Required Construction Work

Construction Work	Area (m ²)
Hangars	100
Design	7
Paving and settling of land	12
Services	200

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 10 Machinery and Equipment

Machinery and Equipment	Quantity	Cost (JD per unit)
Water Treatment Equipment	1	70,000
Sterile rooms / cubic meter	150	350
Water distillation units	1	2,500
Weighing Equipment	2	1,500
Glucose production line	1	488,000
Filling machine	1	5,700
Examination table with lighting	1	3,500
Packaging Machine	1	8,800
Sterilization tools	1	500
Packaging Machine	1	8,800
Quality checking equipments	1	55,000
Filtration Machine	2	1,500
Blender	2	1,500
Forklift	1	12,000

Vehicles and Transportation

The following table shows a list of vehicles and means of transportation required for the project

Table 11 Vehicles and Transportation

Vehicles and Transportation	Quantity	Unit cost (JD)
Pickup truck	2	25,000

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.05% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	30%
Utilities Expenses from Total Revenues	5%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	2.00%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	1.00%

Table 16 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	0%
Income Tax after Deduction	5%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premium (3)	10.12%
Income Tax Rate	5.00%
Beta (4)	1
Risk Premium	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 17 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	577,301
Loans Value	865,952
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	865,952
Loans Percentage	%60
Equity	
Cost of Equity	%16.6
Equity Value	577,301
Equity Percentage	%40
Gross	
Project Value	1,443,253
Proportion of the project	%100
Weighted Average Cost of Capital	%12.05

Capital Expenditures

The estimated cost of the project is 1,443,253JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 18 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	25	5,000	125,000
Cost			125,000

Table 19 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Henger	100		200,000
Design	7		14,000
Paving and settling of land	12		6,000
Facilities	200		10,000
Total			230,000

Table 20 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Water Treatment Equipment	1	70,000	70,000
Sterile rooms / cubic meter	150	350	52,500
Water distillation units	1	2,500	2,500
Weighing Equipment	2	1,500	3,000
Glucose production line	1	488,000	488,000
Filling machine	1	5,700	5,700
Examination table with lighting	1	3,500	3,500
Packaging Machine	1	8,800	8,800
Sterilization tools	1	500	500
Packaging Machine	1	8,800	8,800
Quality checking equipments	1	55,000	55,000
Filtration Machine	2	1,500	3,000
Blender	2	1,500	3,000
Forklift	1	12,000	12,000
Total (JD)			716,300

Table 21 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	25,000
Total			25,000

Table 22 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	2	25,000	50,000
Total			50,000

Table 23 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,500
Transportation	1,500
Marketing	1,500
Training	1,000
Total	5,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	108,968
General and administrative expenses	3	%25	2,926
Marketing expenses	3	%25	1,954
Indirect Salaries	3	%25	46,400
Total			160,248

Table 25 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	125,000	10%	137,500	%9.5
Construction works	230,000	10%	253,000	%17.5
Machinery & Equipment	716,300	10%	787,930	%54.6
Furniture	25,000	10%	27,500	%1.9
Vehicles	50,000	10%	55,000	%3.8
Pre-Operating Expenses	5,500	10%	6,050	%0.4
Working capital	160,248	10%	176,273	%12.2
Total (JD)	1,312,048		1,443,253	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (1,443,253) JD, where 60% of the project will be financed through loans (865,952) JD, and the remaining (40%) through (Equity) with (577,301) JD. The following table summarizes the general structure of the required financing:

Table 26 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	577,301	%40
Loans	865,952	%60
Total	1,443,253	%100
Use of Fund	Value	Ratio
Capital Expenditures	1,260,930	%87.37
Pre-Operating Expenses	6,050	%0.42
Working Capital	176,273	%12.21
Total	1,443,253	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 27 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	234,511	271,740	301,933	332,126	362,320	392,513	422,706	452,900	483,093	513,286
Utilities from	39,085	45,290	50,322	55,354	60,387	65,419	70,451	75,483	80,516	85,548
Maintenance from	7,817	9,058	10,064	11,071	12,077	13,084	14,090	15,097	16,103	17,110
Disposables from	15,634	18,116	20,129	22,142	24,155	26,168	28,180	30,193	32,206	34,219
Direct Salaries	99,200	103,267	107,501	111,909	116,497	121,273	126,246	131,422	136,810	142,419
Others	39,625	44,747	48,995	53,260	57,544	61,846	66,167	70,509	74,873	79,258
Total	435,872	492,218	538,945	585,862	632,979	680,302	727,841	775,604	823,601	871,840

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 28 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	900	927	955	983	1,013	1,043	1,075	1,107	1,140	1,174

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Stationary and Printing	600	618	637	656	675	696	716	738	760	783
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Training	500	515	530	546	563	580	597	615	633	652
Insurance	6,305	6,494	6,689	6,889	7,096	7,309	7,528	7,754	7,987	8,226
Miscellaneous	1,000	1,103	1,136	1,170	1,205	1,241	1,278	1,317	1,356	1,397
Total	11,705	12,128	12,492	12,867	13,253	13,651	14,060	14,482	14,916	15,364

Marketing Expenses

Table 29 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	7,817	9,058	10,064	11,071	12,077	13,084	14,090	15,097	16,103	17,110

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summarizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 30 Expected staff preparation

Job title	No.	Monthly salaries	Total monthly salaries	Total yearly salaries
General manager	1	2,500	2,500	40,000
Production Manager	1	1,500	1,500	24,000
Financial Manager	1	1,750	1,750	28,000
Marketing manager	1	1,500	1,500	24,000
Human Resource Manager	1	1,250	1,250	20,000
Secretaries	1	450	450	7,200
Accountant	2	450	900	14,400
Salesman	2	450	900	14,400
Human Resources Officer	1	450	450	7,200
Guard	1	400	400	6,400
cleaner	1	380	380	6,080
Total Indirect staff	13		9,850	185,600
Technicians	3	400	1,200	19,200
Workers	12	350	4,200	67,200
Driver	2	400	800	12,800
Total direct staff	17		6,200	99,200
Grand Total	30		16,050	284,800

Table 31 Expected monthly salaries (2018-2027)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
General manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Production Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Financial Manager	1,750	1,822	1,896	1,974	2,055	2,139	2,227	2,318	2,413	2,512
Marketing manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Human Resource Manager	1,250	1,301	1,355	1,410	1,468	1,528	1,591	1,656	1,724	1,795
Secretaries	450	468	488	508	528	550	573	596	621	646
Accountant	450	468	488	508	528	550	573	596	621	646
Salesman	450	468	488	508	528	550	573	596	621	646
Human Resources Officer	450	468	488	508	528	550	573	596	621	646
Guard	400	416	433	451	470	489	509	530	552	574
cleaner	380	396	412	429	446	465	484	503	524	546
Direct staff										
Technicians	400	416	433	451	470	489	509	530	552	574
Workers	350	364	379	395	411	428	445	464	483	502
Driver	400	416	433	451	470	489	509	530	552	574

Table 31 Annual Expected Salaries

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
General manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Production Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Financial Manager	28,000	29,148	30,343	31,587	32,882	34,230	35,634	37,095	38,616	40,199
Marketing manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Human Resource Manager	20,000	20,820	21,674	22,562	23,487	24,450	25,453	26,496	27,583	28,714
Secretaries	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Accountant	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Salesman	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Human Resources Officer	7,200	7,495	7,803	8,122	8,455	8,802	9,163	9,539	9,930	10,337
Guard	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Total Indirect	185,600	193,210	201,131	209,378	217,962	226,898	236,201	245,886	255,967	266,462
Direct staff										
Technicians	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Workers	67,200	69,955	72,823	75,809	78,917	82,153	85,521	89,028	92,678	96,477
Driver	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Total Direct	99,200	103,267	107,501	111,909	116,497	121,273	126,246	131,422	136,810	142,419

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Grand Total	284,800	296,477	308,632	321,286	334,459	348,172	362,447	377,307	392,777	408,881

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 32 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	137,500	%0.0	%0.0
Construction Works	253,000	%5.0	%0.0
Machinaries	787,930	%10.0	%2.0
Furniture	27,500	%10.0	%5.0
Vehicles	55,000	%10.0	%0.0
	1,260,930		

Table 33 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,260,930	1,278,064	1,295,581	1,313,492	1,331,807	1,350,536	1,369,690	1,389,279	1,409,316	1,429,811
Total Depreciation Expenses	99,693	101,406	103,158	104,949	106,781	108,654	110,569	112,528	114,532	116,581
Total Accumulated Depreciation	99,693	201,099	304,257	409,207	515,987	624,641	735,210	847,738	962,269	1,078,851
Total Additions	0	17,134	17,518	17,911	18,315	18,729	19,154	19,589	20,036	20,495
Total Net Book Values	1,161,237	1,076,964	991,324	904,286	815,820	725,895	634,480	541,541	447,046	350,960

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 34 Loan Details

Loan Value					865,952
Annual Interest Rate					9.48%
Loans Period					5
Loan Starts at year:					2018
Annual Payment					225,409
Number of Payments					5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				865,952
2019	225,409	82,092	143,317	722,635
2020	225,409	68,506	156,904	565,731
2021	225,409	53,631	171,778	393,953
2022	225,409	37,347	188,063	205,891
2023	225,409	19,518	205,891	0

Income Statement

Table 35 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	781,704	905,800	1,006,444	1,107,088	1,207,733	1,308,377	1,409,021	1,509,666	1,610,310	1,710,955
Gross Operating Revenues	781,704	905,800	1,006,444	1,107,088	1,207,733	1,308,377	1,409,021	1,509,666	1,610,310	1,710,955
Operating Expenses	-435,872	-492,218	-538,945	-585,862	-632,979	-680,302	-727,841	-775,604	-823,601	-871,840
Gross Operating Profit	345,832	413,581	467,499	521,226	574,754	628,075	681,180	734,062	786,710	839,115
<i>Gross Profit Percentage</i>	<i>%44</i>	<i>%46</i>	<i>%46</i>	<i>%47</i>	<i>%48</i>	<i>%48</i>	<i>%48</i>	<i>%49</i>	<i>%49</i>	<i>%49</i>
Salaries and Benefits (Indirect Staff)	-185,600	-193,210	-201,131	-209,378	-217,962	-226,898	-236,201	-245,886	-255,967	-266,462
General and Administraive Expenses	-11,705	-12,128	-12,492	-12,867	-13,253	-13,651	-14,060	-14,482	-14,916	-15,364
Markeing Expenses	-7,817	-9,058	-10,064	-11,071	-12,077	-13,084	-14,090	-15,097	-16,103	-17,110
Pre Operating Expenses	-6,050									
Gross Indirect Expenses	-211,172	-214,396	-223,688	-233,315	-243,292	-253,633	-264,352	-275,464	-286,986	-298,935
Income before Interest, Depreciation and Tax	134,660	199,185	243,811	287,910	331,461	374,442	416,829	458,598	499,723	540,180
Fixed Assets Depreciations	-99,693	-101,406	-103,158	-104,949	-106,781	-108,654	-110,569	-112,528	-114,532	-116,581
Income before Tax and Interests	34,967	97,779	140,653	182,961	224,681	265,788	306,260	346,070	385,192	423,599
Bank Interests	-82,092	-68,506	-53,631	-37,347	-19,518	-0	-0	-0	-0	-0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	-47,125	29,273	87,022	145,614	205,162	265,788	306,260	346,070	385,192	423,599
Income Tax	0	0	-4,351	-7,281	-10,258	-13,289	-15,313	-17,303	-19,260	-21,180
Net Profit	-47,125	29,273	82,671	138,334	194,904	252,499	290,947	328,766	365,932	402,419
<i>Net Profit Percentage</i>	<i>%6-</i>	<i>%3</i>	<i>%8</i>	<i>%12</i>	<i>%16</i>	<i>%19</i>	<i>%21</i>	<i>%22</i>	<i>%23</i>	<i>%24</i>
Compulsory Reserves	0	-2,927	-8,267	-13,833	-19,490	-25,250	-29,095	-32,877	-36,593	-40,242
Retained Earnings	-47,125	-20,779	53,624	178,125	353,538	580,787	842,639	1,138,529	1,467,868	1,830,045

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 36 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	-47,125	29,273	82,671	138,334	194,904	252,499	290,947	328,766	365,932	402,419
Non-monetary adjustments										
Bank benefits	82,092	68,506	53,631	37,347	19,518	0	0	0	0	0
Consumption	99,693	101,406	103,158	104,949	106,781	108,654	110,569	112,528	114,532	116,581
Operating cash flows before changes in working capital	134,660	199,185	239,460	280,630	321,203	361,153	401,516	441,294	480,464	519,000

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Shortage (increase) in inventory	-19,543	-3,102	-2,516	-2,516	-2,516	-2,516	-2,516	-2,516	-2,516	-2,516
(Decrease) increase in accounts payable	33,560	3,186	2,756	2,787	2,821	2,855	2,891	2,928	2,967	3,008
Change in working capital	14,018	84	239	271	304	339	375	412	451	492
Net cash flows (used) from operating activities	148,678	199,269	239,700	280,901	321,508	361,491	401,891	441,706	480,915	519,491
Cash flows from investing activities										
(Purchase) of fixed assets	-1,260,930	-17,134	-17,518	-17,911	-18,315	-18,729	-19,154	-19,589	-20,036	-20,495
Net flows (used) from investment activities	-1,260,930	-17,134	-17,518	-17,911	-18,315	-18,729	-19,154	-19,589	-20,036	-20,495
Cash flows from financing activities										
Equity capital	577,301									
Loan Amortization	-143,317	-156,904	-171,778	-188,063	-205,891	0	0	0	0	0
Bank Interest Rate	-82,092	-68,506	-53,631	-37,347	-19,518	-0	-0	-0	-0	-0
Loans	865,952	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	1,217,844	-225,409	-225,409	-225,409	-225,409	0	0	0	0	0
Net increase (decrease) in cash	105,592	-43,274	-3,227	37,580	77,783	342,762	382,737	422,117	460,878	498,996
Cashflows at the Beginning of Period	0	105,592	62,318	59,091	96,671	174,455	517,217	899,954	1,322,071	1,782,949
Cashflows at the End of Period	105,592	62,318	59,091	96,671	174,455	517,217	899,954	1,322,071	1,782,949	2,281,945

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	105,592	62,318	59,091	96,671	174,455	517,217	899,954	1,322,071	1,782,949	2,281,945
Inventory	19,543	22,645	25,161	27,677	30,193	32,709	35,226	37,742	40,258	42,774
Total assets traded	125,134	84,963	84,252	124,348	204,648	549,927	935,180	1,359,813	1,823,207	2,324,719
Non-current assets										
Fixed assets (net)	1,161,237	1,076,964	991,324	904,286	815,820	725,895	634,480	541,541	447,046	350,960
Total non-current assets	1,161,237	1,076,964	991,324	904,286	815,820	725,895	634,480	541,541	447,046	350,960
Total assets	1,286,371	1,161,927	1,075,575	1,028,634	1,020,468	1,275,822	1,569,659	1,901,354	2,270,253	2,675,679
Liabilities	-	-	-	-	-					
Current liabilities										
Payables	33,560	36,746	39,502	42,289	45,110	47,965	50,856	53,784	56,751	59,759
Remaining amount of Loan	156,904	171,778	188,063	205,891	0-	0-	0-	0-	0-	0
Total current liabilities	190,464	208,524	227,564	248,180	45,110	47,965	50,856	53,784	56,751	59,759
Non-current liabilities										
long term loans	565,731	393,953	205,891	0	0	0	0	0	0	0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total long - term liabilities	565,731	393,953	205,891	0	0	0	0	0	0	0
Total liabilities	756,195	602,478	433,455	248,180	45,110	47,965	50,856	53,784	56,751	59,759
Property rights										
Member Contributions	577,301	577,301	577,301	577,301	577,301	577,301	577,301	577,301	577,301	577,301
Compulsory reserve	0	2,927	11,194	25,028	44,518	69,768	98,863	131,739	168,333	208,574
Profit (loss) round	47,125-	20,779-	53,624	178,125	353,538	580,787	842,639	1,138,529	1,467,868	1,830,045
Total equity	530,176	559,449	642,120	780,454	975,358	1,227,857	1,518,804	1,847,570	2,213,502	2,615,921
Total liabilities and equity	1,286,371	1,161,927	1,075,575	1,028,634	1,020,468	1,275,822	1,569,659	1,901,354	2,270,253	2,675,679

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-1,443,253	154,728	182,136	222,182	262,990	303,193	342,762	382,737	422,117	460,878	498,996	6,445,376
Discount Factor	1.00	0.89	0.80	0.71	0.63	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash	-1,443,253	138,086	145,064	157,926	166,827	171,644	173,175	172,573	169,859	165,510	159,925	2,065,702

Table below (39) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 39 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,443,253	154,728	182,136	222,182	262,990	303,193	342,762	382,737	422,117	460,878	498,996
Project Value Returned	1,443,253	1,288,526	1,106,390	884,208	621,218	318,025	-24,737	-407,474	-829,591	1,290,46	-1,789,465
Payback Period (Year)	5	1	1	1	1	1	0	0	0	0	0

Table 40 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.05
Net Present Value for Cashflows	2,243,038
Payback Period	5
Internal Rate of Return	%25.34

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,243,038 Jordan Dinars considering that the project provides 30 Job Opportunities for the governorate residents.

Table 41 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%25.34	5	12.05%
Revenues declined by 10%	%21.45	7	12.05%
Operating Expenses Increased by 10%	%21.70	6	12.05%